**DO YOU KNOW...?**

One gram of protein on oxidation yields 4 Kcal

## Protein content of foods

The protein contents (range) of various groups of foods are given in **table 10.2**

**Table 10.2:** Protein content (range) of different groups of foods

| Food groups | Protein content g/kg |
|---|---|
| Cereals and millets | 6 -14 |
| Pulses (legumes) dry | 18 – 24 |
| Oilseeds and nuts | 18 – 40 |
| Meat, fish and liver | 18 -20 |
| Eggs | 12 – 14 |
| Milk(fresh)/100ml | 3.5 – 4.0 |
| Milk,(whole) powder | 26 – 28 |
| Milk,(skimmed )powder | 33 - 38 |
| **Vegetables(fresh)** | |
| Leafy | 1 – 4 |
| Roots and tubers | 1 – 1.5 |
| Other vegetables | 1 - 7 |

## 10.5 Functions of proteins

Proteins form a major part of total body structures and they participate in many activities in our body. The major functions of protein in our body is presented in **table 10.3.**

## 10.6 Protein requirements

The important factors affecting the utilization of dietary proteins are the following:

1) **Calorie intake :** For the maximum utilization of dietary proteins, the calorie intake should be adequate. If the calorie intake is inadequate, a part of the dietary protein will be wasted in meeting the energy requirements and the protein need will not be satisfied.

2) **Digestibility co-efficient of proteins:** In the utilization of dietary proteins, a part of the proteins is lost in digestion and in metabolism.

3) **Biological or nutritive value:** The protein in the diet should be derived from different sources such as cereals, pulses, nuts and oilseeds, milk and flesh foods. Since animal proteins possess, in general, a higher nutritive value than vegetable proteins, the diets of children, expectant and nursing mothers should in particular, contain large amounts of proteins derived from milk, eggs and fleshy foods.

The protein requirements depend on age and physiological state of the individual. The ICMR recommended dietary allowance for Indians is shown in **table 10.4.**

## 10.7 Effects of protein deficiency

Diseases due to the deficiency of proteins and calories occur commonly among weaned infants and pre-school children in India and other developing countries.

### 10.7.1 Protein Energy Malnutrition

Protein Energy Malnutrition (PEM) is defined as a range of pathological conditions arising from coincident lack of varying proportions of protein and calorie,



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